

Cal proposal v0.1 — sharp falsifier-test design for the ‘K-types are the particles’ keystone bet

Cal A. Brate (Cal-lane opening contribution; for Lyra refinement + execution)

2026-05-30 Saturday ~09:20 EDT

Sharp falsifier-test design for the “K-types are the particles” bet

0. Why this matters

Per Lyra’s Friday milestone + Keeper’s K-Audit Dictionary First Wave Finding 3: the whole static-taxonomy “DERIVED” rides on a single isolated bet —

The keystone bet: every SM particle state is the substrate-identification of a canonical basis element of $U_v(B_2)$ (and its affine extension $U_v(B_2)^{(1)}$) with the corresponding K-type weight; the K-type’s quantum numbers (σ_{BF} , region, charge, etc.) are derived from the weight structure.

Four of the five static axes are DERIVED-modulo-this-bet (σ_{BF} , region, chirality, particle/anti); charge is LOCATED+constrained-modulo-this-bet; generation is the open mechanism gate. If the bet falls, the whole static dictionary unflips. That’s a lot of weight on one identification.

Per Cal #27 + Keeper Finding 3: every DERIVED cell + headline must carry “modulo keystone bet” — and the standing rule is enforced. But the *test* of the bet itself is the next-order discipline: what would falsify or confirm it directly, not just label its conditionality? Diffuse testing (each dictionary flip independently validates pieces) doesn’t sharply test the bet; it’s better to design a focused falsifier.

This proposal sketches 5 sharp falsifier-test candidates, ranked by sharpness + execution feasibility. Lyra refines + executes; Cal cold-reads results.

1. The bet, made precise (Cal #29 question-shape)

The bet has internal structure worth separating:

- B1 (count claim): at each K-type $V_{(\lambda_1, \lambda_2)}$, the canonical basis dimension equals the count of physical SM particles with that K-type identification. (e.g., at $(1/2, 1/2)$: canonical basis dim should equal the lepton-count = 18 SM-observable or 24 full-Dirac.)
- B2 (quantum-number claim): each canonical basis element’s weight-derived quantum numbers (σ_{BF} from weight parity; charge from $SO(2)$ -weight + GMN; region from Shilov/bulk stabilizer; chirality from J-sign) match a unique SM particle’s quantum numbers.
- B3 (no-ghosts claim): every canonical basis element corresponds to an OBSERVED SM particle — no “ghost” elements with no SM counterpart.
- B4 (no-missing claim): every SM particle has a canonical basis element correspondent — no “missing” SM particle predicted by canonical basis but unobserved.
- B5 (dynamics claim): canonical basis multiplication (Hall product) corresponds to physical fusion/reaction; coproduct to decay; R-matrix to scattering. The dynamic axis rides on this.

Each sub-claim can be falsified independently. Tests below target specific sub-claims.

2. Falsifier candidates (5 tests, sharpness-ranked)

Test T1 — Count test at the lepton K-type (sharpest, executable now)

Target: B1 (count claim).

Design: compute $\dim(\text{canonical basis of } U_v(B_2))$ at weight $(1/2, 1/2)$ for substrate-natural specialization ($v=2$ Hall-Littlewood / field-size; or as appropriate per the affine pin). Compare to 18 (SM-observable lepton states) or 24 (full Dirac with ν_R counted). Mismatch $\rightarrow$ bet fails (at this K-type at least; full count requires testing across K-types).

Sharpness: HIGH. The canonical basis dimension at a specific weight is a STANDARD COMPUTATION in quantum group rep theory; result is forced. If $\dim \neq 18$ (or 24), the bet is broken at the lepton K-type — a clean break. If $\dim = 18$ (or 24), strong confirmation (numerical, but at least the count works).

Execution path: Elie's lane. Probably a few hours' computation. Pin the affine type first (Saturday plan P1.3); then $\dim$ of canonical basis at $(1/2, 1/2)$ is computable from PBW basis + Kashiwara crystal arguments.

Calibration #33 sourcing: the $\dim$ formula for canonical basis at weight λ is a standard quantum group result (Lusztig/Kashiwara), VERIFIED-CITED-able. Apply.

Test T2 — Quantum-number partition test at the adjoint K-type

Target: B2 (quantum-number claim) + B3 (no-ghosts) on the gauge sector.

Design: at K-type $(1, 1)$ (the adjoint, $\dim 10$), the canonical basis elements should partition into electroweak gauge bosons (4: $W^\pm, Z$, photon — though γ might be elsewhere) by their weight-derived quantum numbers. If the 10 elements include items with NO SM gauge-boson counterpart (excess), bet fails B3 (ghost). If SM gauge bosons can't all be accommodated by the 10 (deficit), bet fails B4 (missing) at this K-type.

Sharpness: MEDIUM-HIGH. The 10-dim adjoint accommodating EW gauge bosons + 8 gluons = 12 SM gauge bosons is a known mismatch (the bulk-color $v0.1$ open gate). So this test interacts with the bulk-color question — the test's sharpness depends on resolving where color/gluons sit (bulk per Lyra $v0.1$). If bulk-color resolution moves gluons OUT of the adjoint K-type, the adjoint should carry exactly 4 (EW) — testable.

Execution path: joint with bulk-color $v0.2$. Probably weeks-multi-week as the bulk-color mechanism develops.

Test T3 — Cross-multiplication test (Hall product = physical fusion?)

Target: B5 (dynamics claim).

Design: pick a SPECIFIC known SM vertex (e.g., lepton-photon coupling: $\ell\gamma\ell$ at the QED vertex). Compute the Hall product $(1/2, 1/2) \times (1, 0)$ in $U_v(B_2)$ canonical basis. Result should contain $(1/2, 1/2)$ (the lepton output) with structure-constant matching the QED coupling magnitude (or a substrate-natural ratio thereof). Mismatch on STRUCTURE-CONSTANT shape $\rightarrow$ bet fails B5; mismatch on which K-types appear in the product $\rightarrow$ bet fails B5+ at this vertex.

Sharpness: MEDIUM. The Hall product structure constants at $v=2$ are Mersenne-arithmetic-natural (Elie Toy 3588 territory), so the SHAPE of the structure constants will be computable. The MATCH to SM coupling magnitudes is scheme-dependent (Calibration #33 applies — likely IDENTIFIED-tier, not derived). But the QUALITATIVE structure (which K-types appear in the product) is forced.

Execution path: Elie's lane (Hall product computation) + Lyra's dictionary interpretation. Some weeks.

Test T4 — Forbidden-particle test (look for ghosts)

Target: B3 (no-ghosts).

Design: enumerate ALL canonical basis elements at low K-types (say, all K-types with $\dim \leq 20$). For each, compute weight-derived quantum numbers. Check whether each has an SM counterpart. ANY canonical basis element with weight-derived quantum numbers NOT MATCHING any SM particle (e.g., a charge that isn't 0,

$\pm 1/3, \pm 2/3, \pm 1$; or a chirality combination not in SM) $\rightarrow$ “ghost” $\rightarrow$ either prediction of an unobserved particle (interesting!) or falsifier of the bet (if observed-exclusion is firm).

Sharpness: HIGH for finding ghosts (clean enumeration). MEDIUM for the falsification value (depends on whether observed-exclusion is firm enough — SM is well-tested at low energies, so any low-K-type ghost should already have been observed if real).

Execution path: Elie’s lane (enumerate canonical basis at low K-types) + Cal cold-read on the ghost dispositions. Days to weeks.

Test T5 — Five-Absence falsifier propagation

Target: framework-wide (not just keystone, but the bet would help propagate).

Design: the substrate predicts NO 4th generation, NO sterile neutrinos, NO GUT, NO SUSY, NO monopoles, NO proton decay (per K65 Five-Absence Predictions Set). The keystone bet says these absences come from the canonical basis structure (no canonical basis elements at the relevant K-types). Each Five-Absence prediction is independently falsifiable by experiment. Observation of any $\rightarrow$ falsifier of substrate framework.

Sharpness: HIGH for the absence predictions; experimental timescales are decades (these are SM/BSM searches).

Execution path: standing experimental falsifiers; not Cal-actionable; just track.

3. Test priority + sequence

Test	Sharpness	Execution	Priority
T1 (count at lepton K-type)	HIGH	hours-days (post-affine-pin)	TOP — execute first
T4 (forbidden particles / ghosts)	HIGH	days-weeks	second
T3 (cross-multiplication)	MEDIUM	weeks	third
T2 (gauge partition)	MEDIUM-HIGH	weeks (joint with bulk-color)	fourth
T5 (Five-Absence)	HIGH at observation, experimental timescale	decades	standing

T1 is the cleanest falsifier-test executable now. The count at the lepton K-type should match 18 (or 24); if it doesn’t, the bet falls cleanly. If it does, the bet survives this specific check (still doesn’t prove other K-types).

4. Discipline framing

Per Cal #27 forward-derivation: these tests verify whether the keystone bet’s PREDICTED canonical basis structure MATCHES the SM particle structure (a structural-correspondence test). If matches $\rightarrow$ bet survives (modulo independence of routes); if mismatches $\rightarrow$ bet falls (clean falsification).

Per Cal #29 question-shape: the tests target SPECIFIC sub-claims (B1-B5) — not “does the bet work?” (too diffuse) but “does the canonical basis dim at this K-type match the SM count?” (specific). Each test produces a forced YES/NO.

Per Cal #32 hard-constraint: the tests use STRUCTURAL CONSTRAINTS that the bet must satisfy (canonical basis dim formulas, weight-derived quantum-number consistency, Hall product structure). Not value-match (e.g., not matching specific coupling magnitudes — those are scheme-dependent IDENTIFIED tier).

Per Calibration #33: each test’s input formulas (canonical basis dim, Hall structure constants, weight derivations) should be VERIFIED-CITED at execution time (Lusztig/Kashiwara for canonical basis dim; standard rep theory for weights; Ringel-Hall for Hall structure constants). “Scope is part of the value” — check the formulas cover the $v=2$ Hall-Littlewood corner specifically (the substrate’s specialization point per Cal #155 A1 corrections).

5. What this proposal IS and ISN'T

IS: a Cal-side opening sketch of 5 sharp falsifier candidates for the keystone bet, with sharpness ratings + execution paths. Cal's design contribution to Saturday plan item 4.5.

ISN'T: a derivation or a result. The tests are designs; running them is Lyra (refines) + Elie (computes) + Cal (cold-reads results).

6. Routing

- Lyra (joint with Cal per Keeper plan): refine the test designs; especially T1 (count test) — is the canonical basis dim formula at substrate specialization ($v=2$) computable cleanly, and is the comparison to 18 (SM) or 24 (Dirac) the right target?
- Elie: T1 (count) + T4 (ghosts) + T3 (cross-mult) are Lie/Hall computations in his lane.
- Keeper: tier-gate the test designs + the eventual results.
- Grace: T5 (Five-Absence) catalog-tracks experimental status.
- Cal: cold-read each test's results when executed; apply Calibration #33 sourcing + Cal #32 hard-constraint discipline.

— Cal A. Brate, Saturday 2026-05-30 ~09:21 EDT. Sharp falsifier-test design proposal v0.1 for the “K-types are the particles” keystone bet. 5 tests, sharpness-ranked; T1 (count at lepton K-type) is the cleanest executable-now falsifier. Cal-side opening contribution; Lyra refines + executes. The bet currently carries the whole static taxonomy DERIVED-modulo-keystone; a sharp focused test of it (not diffuse) is the highest-leverage falsifier-design Cal can offer.

Addendum (Saturday 2026-05-30 ~09:30 EDT) — T1 target must be PINNED A-PRIORI (Keeper note absorbed)

Per Keeper's response: “the ‘18 vs 24’ framing depends on how we count the SM-observable spinor count (per-generation? including handedness? CKM-correlated?) — the cleanest version pins those choices a-priori so the test can't be tuned post-hoc.”

This is the Cal #29 question-shape discipline applied to the test design itself — and Keeper is right. A test where the comparison target is tunable post-hoc isn't a sharp falsifier; it's a back-fit opportunity. Refining T1:

T1 — Count test at lepton K-type $V_{(1/2,1/2)}$, PINNED a-priori:

Before the canonical-basis-dim computation runs, the SM-side count must be fixed in writing with explicit choices on all degrees of freedom. Specifically:

T1-pin (declare upfront, before computing):

Choice	Option A (SM-observable)	Option B (full-Dirac substrate-natural)	Cal recommendation
Right-handed neutrinos?	NO (SM convention)	YES (Dirac partner, substrate-natural)	B — substrate has Dirac neutrinos (Paper 108 correction); ν_R is a substrate particle even if not observed in low-energy SM
Left/right chirality counted separately?	YES (2 per charged lepton $\times$ 1 for ν_L only)	YES (2 per charged + 2 per neutrino if Dirac)	YES both — chirality is a substrate-derived axis (L3); each chirality state is a distinct substrate particle

Choice	Option A (SM-observable)	Option B (full-Dirac substrate-natural)	Cal recommendation
Antiparticles counted?	YES (counted as separate states)	YES	YES — particle/anti is a substrate-derived axis (engine v0.4 E/F); each antiparticle is a distinct substrate particle
Generations counted?	YES (3 generations $\times$ all of the above)	YES	YES — generation is indexed (open per #414) but counted
CKM-mixing-correlated?	NO (count in mass basis)	NO (count substrate-natural K-type basis)	NO — the K-type is the substrate identification; mixing across K-types is a separate dynamics question

Recommended T1-target (Cal lean): Option B = 24 — 3 generations $\times$ (2 charged-lepton chiralities $\times$ 2 particle/anti + 2 neutrino chiralities $\times$ 2 particle/anti) = $3 \times (4 + 4) = 24$. This matches Dirac-neutrino substrate-naturalness + counts every substrate-distinguished state.

Option A = 18 follows if (and only if) the SM “no ν_R observed” convention is the substrate-side truth — but that contradicts the substrate’s Dirac-neutrino structure (per Paper 108). So Option B is the substrate-consistent target.

T1-result interpretation (declared upfront): - $\dim(\text{canonical basis at } V_{(1/2,1/2)}) = 24 \rightarrow$ bet survives B1 at the lepton K-type - $\dim = 18 \rightarrow$ bet’s substrate framework needs to explain why ν_R isn’t in canonical basis (Dirac vs Majorana tension) - $\dim = \text{anything else} \rightarrow$ bet falls at B1 (the count is what it is; the framework can’t tune to match) - $\dim$ depends on representation choice / basis convention $\rightarrow$ recheck the QG computation (this is the Cal #32 / Calibration #33 sourcing check — pin to Lusztig/Kashiwara conventions a-priori)

Why this matters (Cal #29 reflexive): with the count-target PINNED upfront, the test produces a forced YES/NO; without pinning, “the test passed” could quietly mean “we picked the SM-side count to match the canonical basis dim we got.” Keeper’s point catches the back-fit risk before it can land. Good cross-CI discipline.

Updated T1 priority: still TOP (sharpness HIGH; executable now after the affine pin); now with target = 24 declared a-priori; Calibration #33 source-pin Lusztig/Kashiwara canonical basis conventions before computing.

— Cal addendum, Saturday 2026-05-30 ~09:30 EDT, absorbing Keeper’s “pin a-priori” note. T1 target = 24 (full Dirac with ν_R) declared upfront; the test now produces a forced YES/NO without post-hoc tuning. Cal #29 question-shape discipline applied to the test design itself.